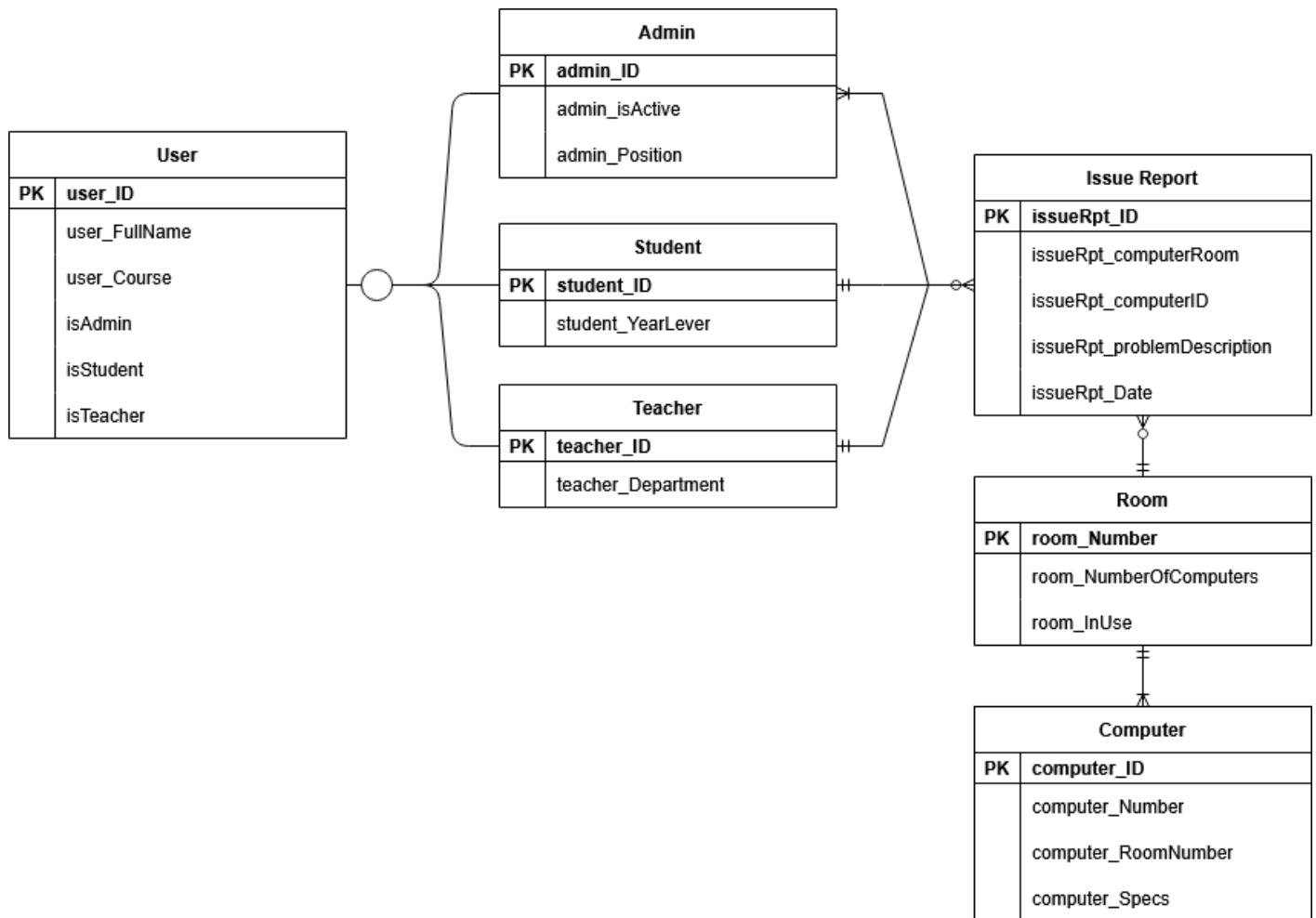


Revised Entity Relationship Diagram

Metadata

Entity 1: User (Parent Table)

Field Name	Description	Data Type	Domain	isNull	isUnique
user_ID	Primary key for the user	Integer	Auto-increment	No	Yes
user_FullName	Full name of the user	String	Max 100 characters	No	No
user_Course	Academic course or program	String	Valid university courses	Yes	No
isAdmin	Flag for Admin subtype	Boolean	0 (False), 1 (True)	No	No
isStudent	Flag for Student subtype	Boolean	0 (False), 1 (True)	No	No
isTeacher	Flag for Teacher subtype	Boolean	0 (False), 1 (True)	No	No

Entity 2: Admin (Subtype of User)

Field Name	Description	Data Type	Domain	isNull	isUnique
admin_ID	PK/FK linked to User Table	Integer	Existing user_IDs	No	Yes
admin_isActive	Status of admin account	Boolean	0 (Inactive), 1 (Active)	No	No
admin_Position	Job title/Role	String	Staff, Tech, Supervisor	No	No

Entity 3: Student (Subtype of User)

Field Name	Description	Data Type	Domain	isNull	isUnique
student_ID	PK/FK linked to User Table	Integer	Existing user_IDs	No	Yes
student_YearLevel	Current year level	Integer	1, 2, 3, 4, 5	No	No

Entity 4: Teacher (Subtype of User)

Field Name	Description	Data Type	Domain	isNull	isUnique
teacher_ID	PK/FK linked to User Table	Integer	Existing user_IDs	No	Yes
teacher_Department	Assigned faculty/dept	String	IT, CS, Eng, etc.	No	No

Entity 5: Room

Field Name	Description	Data Type	Domain	isNull	isUnique
room_Number	Primary ID for the room	String	e.g., "Room 301"	No	Yes
room_NumberOfComputers	Total PC count in room	Integer	Value > 0	No	No
room_InUse	Current availability status	Boolean	0 (No), 1 (Yes)	No	No

Entity 6: Computer

Field Name	Description	Data Type	Domain	isNull	isUnique
computer_ID	Primary ID for the hardware	Integer	Auto-increment	No	Yes
computer_Number	Physical tag number	String	e.g., "PC-2026-01"	No	Yes
computer_RoomNumber	FK linked to Room Table	String	Existing room_Numbers	No	No
computer_Specs	Hardware specifications	Text	CPU, RAM, GPU info	No	No

Entity 7: Issue Report

Field Name	Description	Data Type	Domain	isNull	isUnique
issueRpt_ID	Primary ID for the report	Integer	Auto-increment	No	Yes
issueRpt_computerRoom	FK linked to Room Table	String	Existing room_Numbers	No	No
issueRpt_computerID	FK linked to Computer Table	Integer	Existing computer_IDs	No	No

issueRpt_problemDescription	Detailed maintenance info	Text	Min 10 characters	No	No
issueRpt_Date	Timestamp of report	Date/Time	Current System Date	No	No

Database Schema / Relations

User (user_ID, user_FullName, user_Course, isAdmin, isStudent, isTeacher)

Admin (admin_ID, admin_isActive, admin_Position)

Student (student_ID, student_YearLever)

Teacher (teacher_ID, teacher_Department)

Issue Report (issueRpt_ID, issueRpt_computerRoom, issueRpt_computerID, issueRpt_problemDescription, issueRpt_Date)

Room (room_Number, room_NumberOfComputers, room_InUse)

Computer (computer_ID, computer_Number, computer_RoomNumber, computer_Specs)

Revised Business Rules

1. Every computer must be assigned a unique computer_ID and be logically associated with a room_Number within the Room table.
2. The computer_Specs field must contain detailed technical information, including the processor (CPU), storage capacity, and graphics card (GPU).
3. Users identified as Admins, Students, or Teachers (Subtypes of the User entity) are all permitted to initiate an Issue Report.
4. An Issue Report cannot exist unless it is linked to a valid room_Number and an existing computer_ID from the inventory.
5. Every Issue Report must contain a issueRpt_problemDescription to ensure the technical staff understands the nature of the malfunction.
6. The system must capture the current date and time in the issueRpt_Date field automatically upon the submission of any new report.
7. The system must allow users to filter and find equipment using the computer_Number, computer_ID, or the specific room_Number.
8. Only Users with the isAdmin flag set to true (and recorded in the Admin table) are authorized to verify repairs and mark issues as resolved.
9. When a room or computer is taken out of service (monitored via room_InUse or admin_isActive), the associated Issue Report records must be preserved for audit purposes.
10. The system must generate a dashboard using data from Room, Computer, and Issue Report to display total inventory counts, pending maintenance requests, and the university's overall resolution percentage.

Revised Figma Prototype link

<https://www.figma.com/design/yOu1xQO9j4Xe1cVqvkwUZZ/IM-Figma-Model?node-id=0-1&t=KB6Wl7hkpDXV9nIW-1>